



**Department of Computer Science and
Engineering
National Institute of Technology Karnataka,
Surathkal – 575025**

Course Name : Compiler Design Lab

Course Code : CS304

Lab Number : Week 6 (S2)

Experiment Title : **Conditional and Iterative Statement**

Question 1: Write a YACC program to recognize and validate conditional control structures in a C-like language. The parser should recognize the following statements:

1. if statement
2. Nested if statements

The following general syntax should be supported:

if (condition) statement;

The parser should display “**Valid Statement**” for syntactically correct statements and an appropriate **syntax error message** for invalid statements.

Sample valid inputs:

if (a > b) x = 10;

Question 2:

Write a YACC program to recognize and validate iterative control structures in a C-like language. The parser should recognize:

1. for statement
2. Nested looping statements

The following general syntax should be supported:

for (initialization; condition; increment) statement;

The parser should report an appropriate error for missing parentheses, semicolons, or invalid expressions.

Sample valid inputs:

for (i = 0; i < 10; i = i + 1) x = x + i;

Code and Results Submission Deadline : Between 11:30 – 11:45 AM on the Lab Day
